Simul Ahmed

Embedded Systems & PCB Design Engineer

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PROFILE

Embedded systems and PCB design engineer with full-cycle hardware experience, from schematic and layout in Altium Designer to fabrication outputs and production support. Core R&D contributor to a commercial power-control product with 2,000+ units deployed, designer of a four-board industrial edge system for a contract client, and member of the team behind a healthcare IoT product. Altium-certified, with firmware-aware layout experience on STM32, ESP32 and ATmega. Final-year B.Sc. student at KUET, graduating December 2027.

WORK EXPERIENCE

Caretron — Embedded System Engineer | Dhaka, Bangladesh (Hybrid) *Aug 2025 – Present*

- Member of the team that developed Kairon, a digital medication reminder released in Digital and Smart IoT variants.
- Contributed to R&D, design and inspection, working in Altium Designer and SolidWorks.

FabLab KUET — Student Operator | Khulna, Bangladesh *Jul 2026 – Present*

- Run day-to-day lab operations, maintain equipment and support student projects.
- Lead hands-on workshops and training sessions on PCB design and fabrication.

EasyEDA — Official Campus Ambassador | KUET, Khulna *Jul 2026 – Present*

- Represent EasyEDA at KUET; course instructor for the official EasyEDA PCB training camp (Oct 2026).

Transform AI — PCB Designer (Project-based Contract) | Khulna, Bangladesh *May 2026 – Aug 2026*

- Designed the EDGE NODE, a four-board stacked assembly (main controller, power, relay interface and SBC carrier), with outlines, connectors and mounting verified together in 3D.
- Designed isolated Modbus, isolated RS232 and user-access boards; released the RS232 board for fabrication with Gerber, BOM and pick-and-place outputs.
- Wrote a power study comparing regulator ICs and direct battery powering for an IP camera system.

AuraXIoT — Junior Embedded System Engineer | Dhaka, Bangladesh *Apr 2025 – Jan 2026*

- Core R&D contributor to an Automatic Transfer Switch (ATS) product with 2,000+ units deployed in the field.
- Contributed to end-to-end hardware development: schematic design, PCB layout, enclosure development and BOM preparation.
- Designed for local manufacturing constraints in Bangladesh: PCB fabrication, assembly tolerances and injection-molded enclosures.
- Supported production testing and pre-delivery validation to ensure unit-level reliability.

FabLab KUET — Equipment Restoration Contributor | Khulna, Bangladesh *Oct 2024 – Feb 2025*

- Restored a non-functional Cirqoid CNC PCB mill by diagnosing faults, modifying the bed and building a custom G-code workflow around its software limits.
- Achieved consistent 0.15 mm trace and clearance, restoring in-house PCB prototyping for the lab.

Firefly High Tech — Project Trainee | Khulna, Bangladesh *Aug 2023 – Oct 2025*

- Helped develop and debug 30+ embedded and electronics projects for KUET students, covering sensors, microcontrollers and power systems.

Dept. of Energy Science and Engineering, KUET — Volunteer Research Contributor | Khulna, Bangladesh *Jan 2024 – Oct 2024*

- Built the experimental setup for geothermal research, installing thermocouples in underground test columns, and collected data across multiple configurations.

PRODUCTS AND PROJECTS

Commercial products

Kairon, Digital Medication Reminder (Caretron) | Altium Designer, SolidWorks *2026*

Healthcare product by Caretron: a three-drawer medication reminder for morning, noon and night doses, released in Digital and Smart IoT variants. Kairon is a team product; I was a member of the development team, contributing to R&D, design and inspection.
Portfolio: simul-ahmed.vercel.app/projects/kairon

ATS Power Control System (AuraXIoT) | Altium Designer, Proteus, SolidWorks *Jan 2026 – Apr 2026*

Commercial power-control unit for automatic mains-to-IPS switching in small backup systems (up to ~200 W). Solved instability during grid restoration and protected the IPS charging system. Designed the full schematic and double-sided production PCB; 2,000+ units manufactured.
Portfolio: simul-ahmed.vercel.app/projects/ats-power-control

Projects

Line-Following Robot: System Identification and PID | STM32F411, MATLAB, Altium Designer *Sep 2024 – Apr 2026*

Four generations of one robot, from a milled STM32F411 board to a research platform. Identified a second-order discrete-time plant model from Bluetooth telemetry (56.1% fit) and tuned the PID in MATLAB; under 5% overshoot on hardware.
Portfolio: simul-ahmed.vercel.app/projects/line-following-robot | GitHub: github.com/simulahmed003/System-Identification-and-PID-Control-Implementation-on-LFR

Cirqoid CNC Controller Redesign | GRBL, stepper drivers *Jun 2026 – Present*
Designing a replacement GRBL controller for a PCB mill whose manufacturer no longer exists: four stepper drivers, limit switches and isolation, verified against the original hardware. Reverse-engineered the BLDC spindle driver; first prototype built.
Portfolio: simul-ahmed.vercel.app/projects/cirqoid-cnc | GitHub: github.com/simulahmed003/grbl-custom

IoT Water Temperature Controller | ESP32, Altium Designer *Jan 2025 – Feb 2025*
ESP32 temperature monitor and relay heater control with on-board 12 V to 5 V regulation, taken from a module-based prototype to an integrated double-sided board.
Portfolio: simul-ahmed.vercel.app/projects/iot-water-temperature-controller | GitHub: github.com/simulahmed003/IOT-Water-Temp-Controller

BCD Digital Clock | Altium Designer, Proteus *Jun 2025 – Jul 2025*
Fully discrete HH:MM:SS clock with no microcontroller, built from 74HC BCD counters and logic gates; verified in Proteus and laid out in Altium.
Portfolio: simul-ahmed.vercel.app/projects/bcd-digital-clock | GitHub: github.com/simulahmed003/BCD-Clock

5×5×5 LED Cube Panel and Driver Board | Altium Designer, Arduino Nano *Apr 2025 – May 2025*
Modular LED cube with layered 5×5 matrices and a shift-register driver board for synchronised pattern rendering.
Portfolio: simul-ahmed.vercel.app/projects/led-cube | GitHub: github.com/simulahmed003/5x5x5-LED-Cube

EDUCATION

Khulna University of Engineering & Technology (KUET) *Jan 2023 – Present*
B.Sc. Engineering, Energy Science and Engineering. Expected graduation: December 2027.
Thesis (ongoing): experimental comparison of P&O and Incremental Conductance MPPT on a custom PV battery charger.

Police Lines School and College, Kushtia *Jan 2019 – Jan 2022*
HSC, Science

CERTIFICATIONS

Altium Skilled Engineer (Credly badge), Altium *Oct 2026*
Altium Workspace User Management, Altium (ID: afj4crzof5) *Oct 2026*
Basic Embedded Systems Design, Altium (ID: zswco0vxuy) *Sep 2026*
Altium Designer Essentials, On Demand, Altium (ID: bgxkfw5e1p) *Jul 2026*
Altium Emerging Engineer (Credly badge), Altium *Jul 2026*

TRAINING

SARBS Communication Ltd., Dhaka | Industrial Attachment: Product-level PCB Design in Altium Designer *Feb 2025 – Mar 2025*
- Analysed commercial schematics and built a complete thermocouple data logger schematic from modular open-source hardware blocks.

SARBS Communication Ltd., Dhaka | Industrial Attachment: PCB Fabrication (CNC Milling and Chemical Etching) *Oct 2024*
- Designed, milled and assembled a temperature-controlled fan controller, and etched a compact 3.3 V regulator board.

Orion Power Rupsha Ltd., Khulna | Industrial Attachment: Power Plant Operations *Nov 2025 – Dec 2025*
- Observed power generation, electrical distribution and operational safety procedures with the Operation Department.

ACHIEVEMENTS

2nd Place, Mars Autonomous Rover Rally Design Competition *Dec 2025*
Team Durbar (KUET), 2nd internationally. Organised by the NASA Texas Space Grant Consortium and Texas A&M University.

Top 10 Finalist, VLSITHON 3.0 *Sep 2026*
National chip-design competition by Ulkasemi; grand finalist with team Magical Musketeer.

Bangladesh Physics Olympiad, Regional Round (Category C) *2018, 2019*
7th position (2019) and 9th position (2018), Kushtia region; national round participant both years.

SKILLS

PCB Design	Altium Designer (certified), KiCad, EasyEDA: schematic capture, double-sided layout, multi-board assemblies, isolated and mixed-signal layout, fabrication outputs
Embedded	STM32, ESP32, ATmega (AVR), Arduino; firmware-aware layout decisions; MQTT, Bluetooth
Fabrication & CAM	CNC PCB milling, chemical etching, 3D printing, laser engraving; FlatCAM, Candle, bCNC, custom G-code
Simulation & Control	Proteus, MATLAB / Simulink, system identification, PID design
Mechanical	SolidWorks (enclosure design)
Programming	C / C++, Python, MATLAB
Sourcing	LCSC / JLCPCB and the Bangladeshi local market; cost-optimised, manufacture-ready BOMs
Languages	Bangla (native), English

REFERENCES

Dr. Md. Hasan Ali, Professor, Dept. of Energy Science and Engineering, KUET **Omar Faruk Badhon**, Embedded Systems Engineer, SARBS
Contact details available on request.